FEM - TẤM

REISSNER\MINDLIN

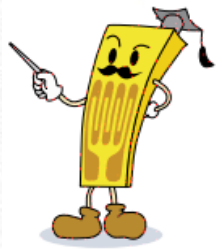
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Tp. HCM 22-01-2013

Nội dung



- Dạng yếu biến phân của tấm Reissner\Mindlin
- Quy ước dấu trong tấm Reissner\Mindlin
- FEM – T3
- FEM – Q4



Nội dung



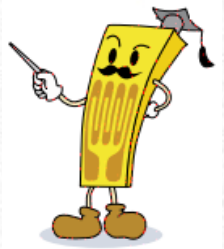
- phương trình dạng yếu biến phân của tấm Reissner\Mindlin

Phương trình cân bằng tĩnh học

$$\int_{\Omega} \delta\boldsymbol{\kappa}^{T} \mathbf{D}^{b} \boldsymbol{\kappa} d\Omega + \int_{\Omega} \delta\boldsymbol{\gamma}^{T} \mathbf{D}^{s} \boldsymbol{\gamma} d\Omega = \int_{\Omega} \delta\mathbf{u}^{T} \mathbf{b} d\Omega$$

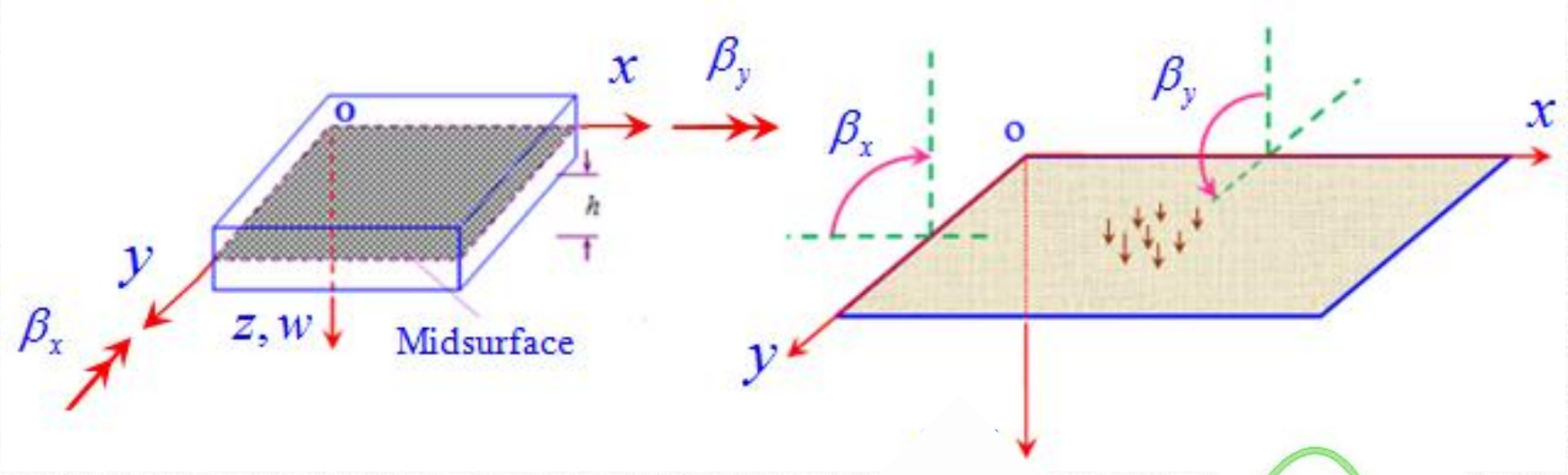
Phương trình cân bằng cho bài toán phân tích dao động tự do

$$\int_{\Omega} \delta\boldsymbol{\kappa}^{T} \mathbf{D}^{b} \boldsymbol{\kappa} d\Omega + \int_{\Omega} \delta\boldsymbol{\gamma}^{T} \mathbf{D}^{s} \boldsymbol{\gamma} d\Omega + \int_{\Omega} \delta\mathbf{u}^{T} \mathbf{m}\ddot{\mathbf{u}} d\Omega = 0$$



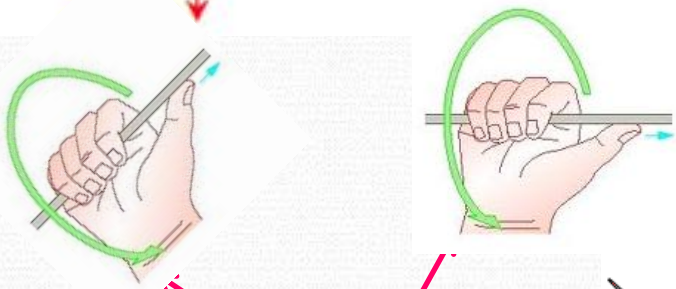
Lý thuyết tấm

➢ Quy ước dấu trong tấm Reissner\Mindlin

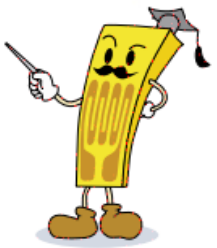


- β_y là góc quay xung quanh trục x
- β_x là góc quay xung quanh trục y

➔ Xác định theo quy tắc bàn tay phải



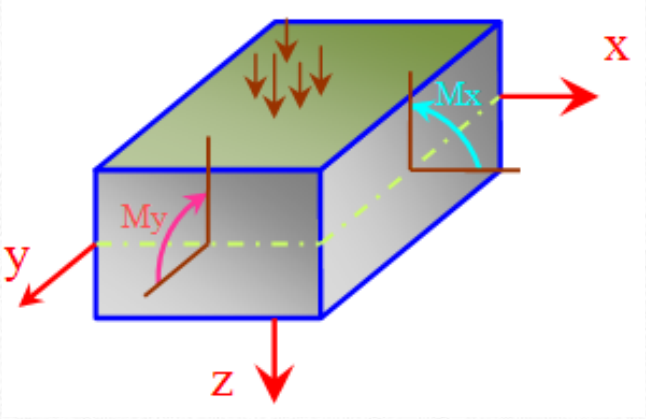
Chiều quay của β theo độ võng

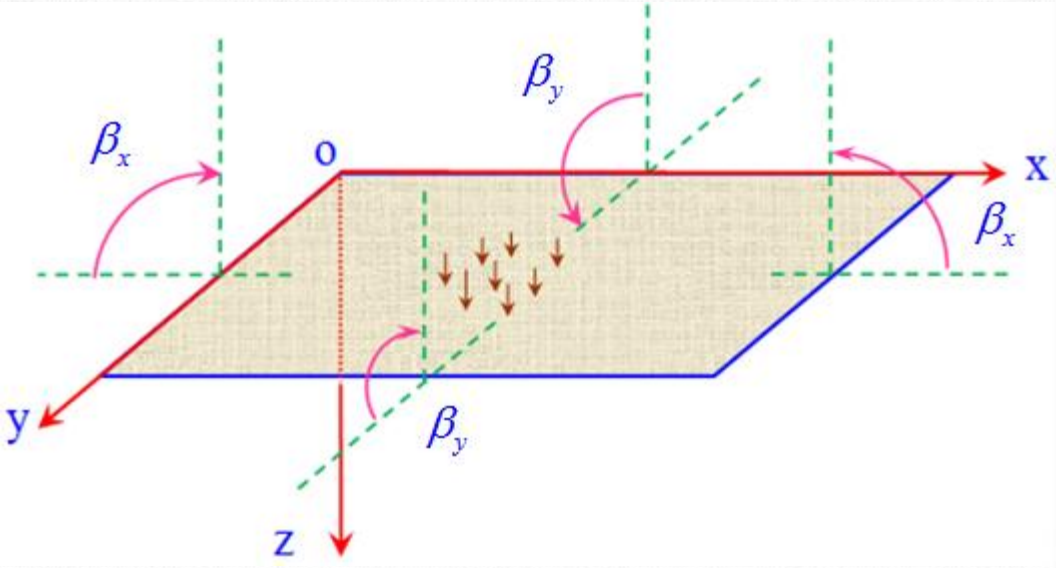


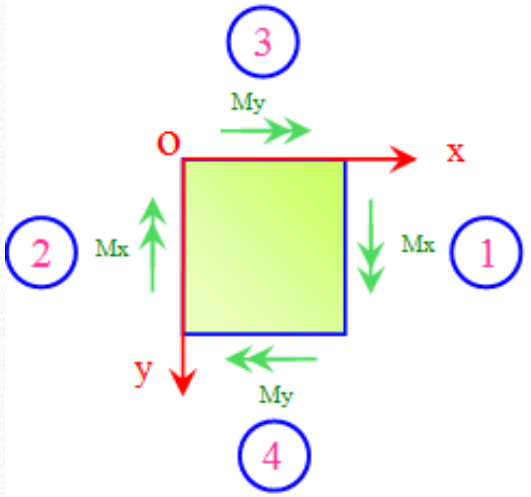
Lý thuyết tấm



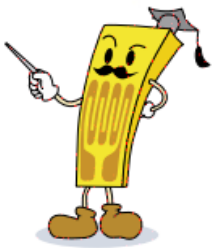
➢ Quy ước dấu trong tấm Reissner\Mindlin





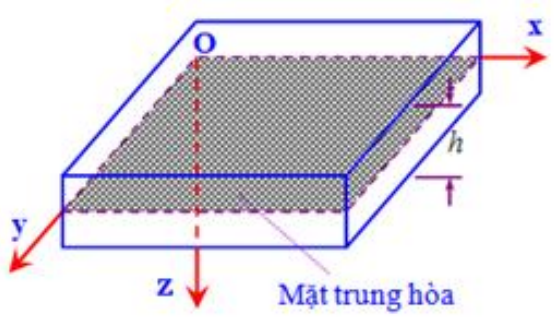


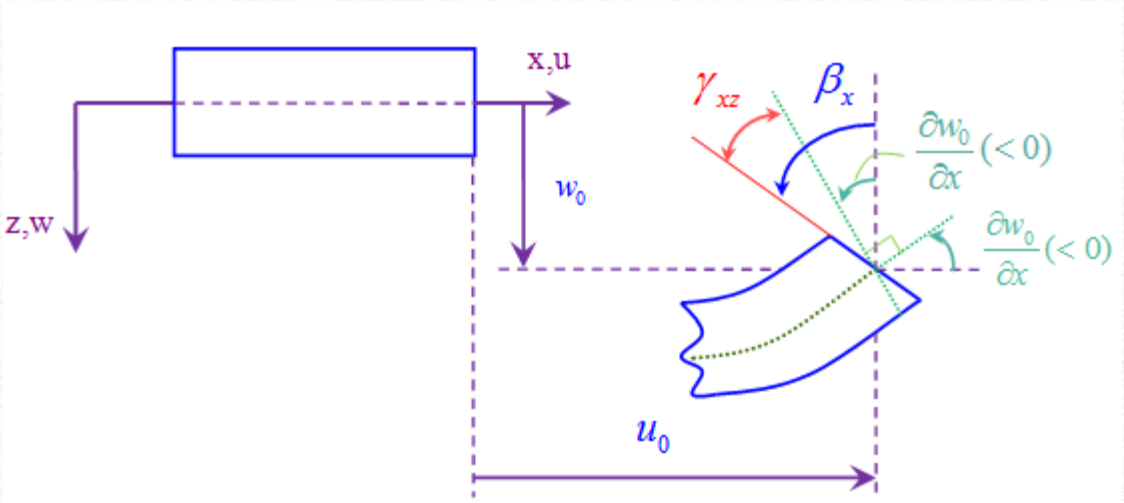
○ Chiều quay của β chính là chiều của moment



Lý thuyết tấm

➢ Quy ước dấu trong tấm Reissner\Mindlin





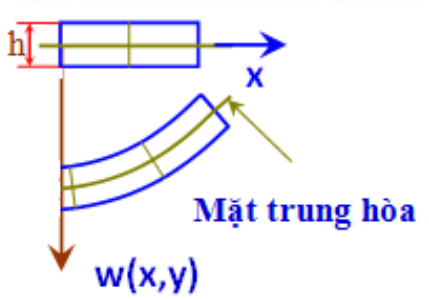
$\frac{\partial w_0}{\partial x}(<0)$ Góc xoay do biến dạng uốn

γ_{xz} Góc xoay do biến dạng cắt

β_x Góc xoay tổng

Theo trục x,

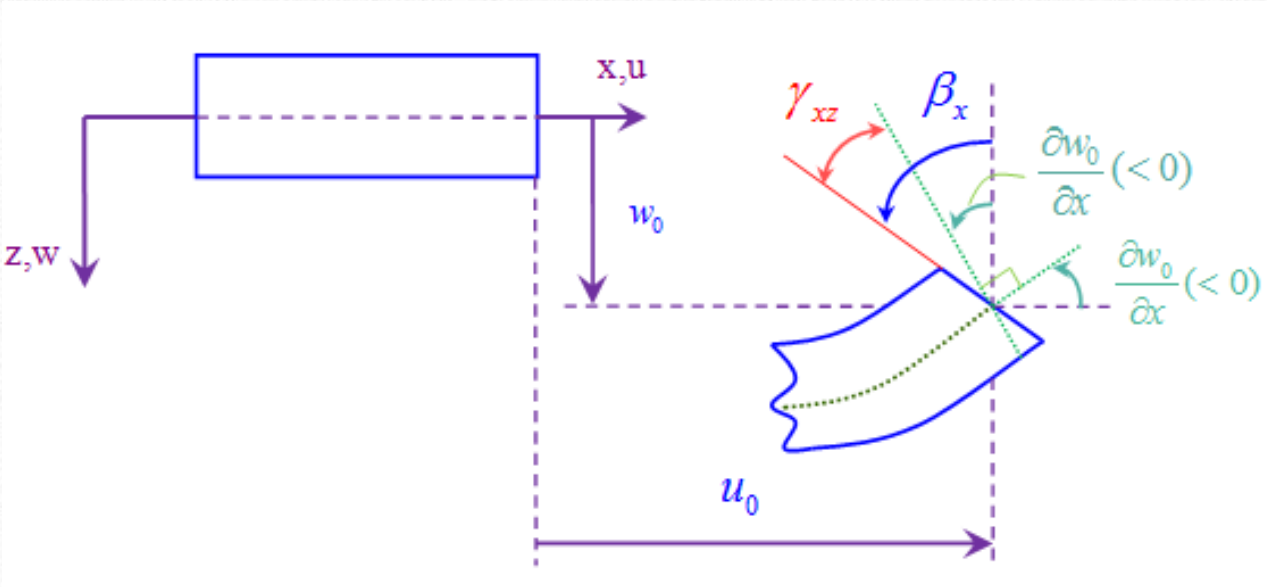
w giảm



$$\frac{\partial w_o}{\partial x} = \lim_{\Delta x \to 0} \frac{w_o(x+\Delta x) - w_0(x)}{\Delta x} < 0$$

Lý thuyết tấm

➢ Quy ước dấu trong tấm Reissner\Mindlin



○ Góc xoay do biến dạng cắt gây ra

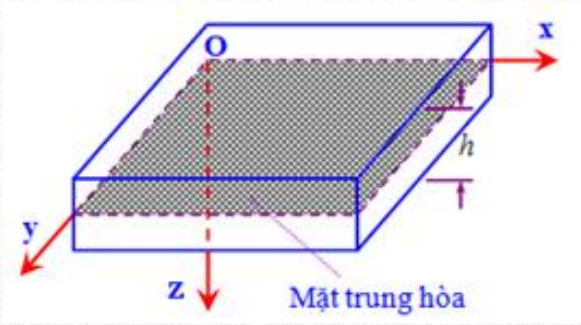
$$\gamma_{xz} = \frac{\partial w_0}{\partial x} + \beta_x = w_x' + \beta_x$$

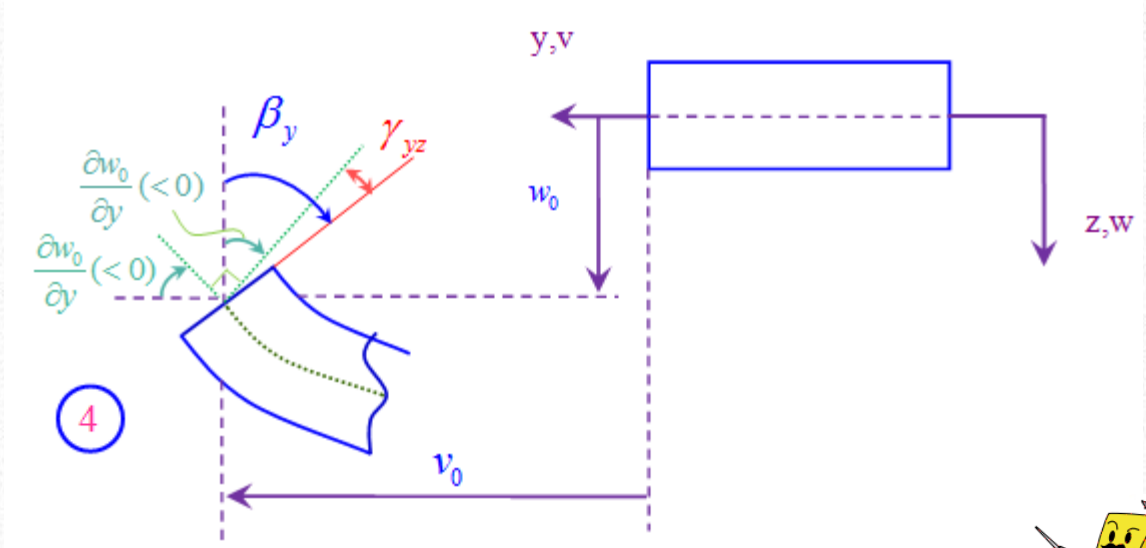
○ Chuyển vị theo phương x

$$u = u_0 + z\beta_x$$

Lý thuyết tấm

➢ Quy ước dấu trong tấm Reissner\Mindlin





$\frac{\partial w_0}{\partial y}(<0)$ Góc xoay do biến dạng uốn

γ_{yz} Góc xoay do biến dạng cắt

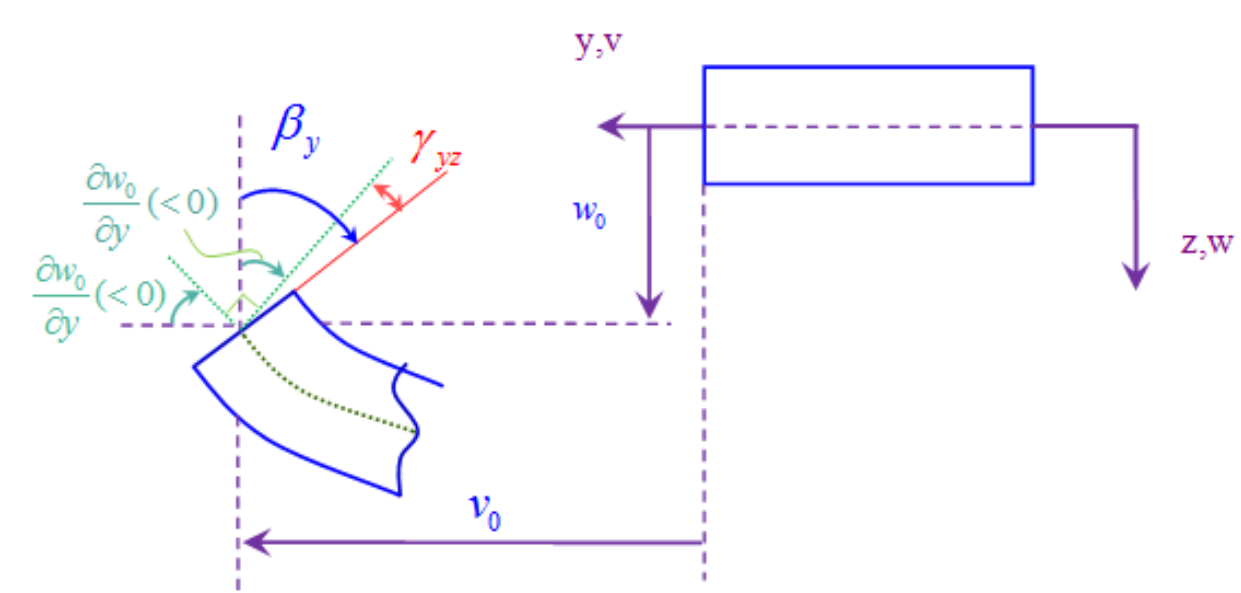
β_y Góc xoay tổng

Theo trục y, *w* giảm

$$\frac{\partial w_o}{\partial y} = \lim_{\Delta y \to 0} \frac{w_o(y+\Delta y) - w_0(y)}{\Delta y} < 0$$

Lý thuyết tấm

➢ Quy ước dấu trong tấm Reissner\Mindlin



○ Góc xoay do biến dạng cắt gây ra

$$\gamma_{yz} = \frac{\partial w_0}{\partial y} + \beta_y = w'_y + \beta_y$$

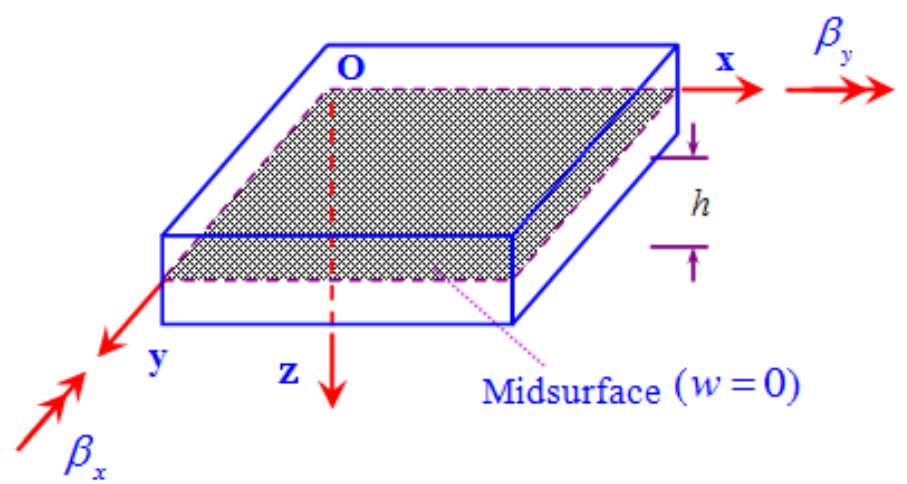
○ Chuyển vị theo phương x

$$v = v_0 + z\beta_y$$

➢ Thành lập công thức Bb

○ Chuyển vị theo 3 phương x – y - z

$$u = u_0 + z\beta_x$$
$$v = v_0 + z\beta_y$$
$$w = w_0$$



○ Biến xấp xỉ theo thứ tự z – y - x

$$\mathbf{U}_\theta = \begin{bmatrix} w \\ \beta_x \\ \beta_y \end{bmatrix} = \begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix} \begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}$$

$w_i \Rightarrow$ Trục z

$\beta_{xi} \Rightarrow$ Trục y

$\beta_{yi} \Rightarrow$ Trục x

$$\mathbf{U}_\beta = \mathbf{N}_\beta \mathbf{d}_\beta$$



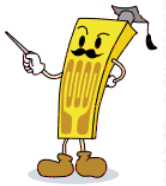


➢ Thành lập công thức Bb

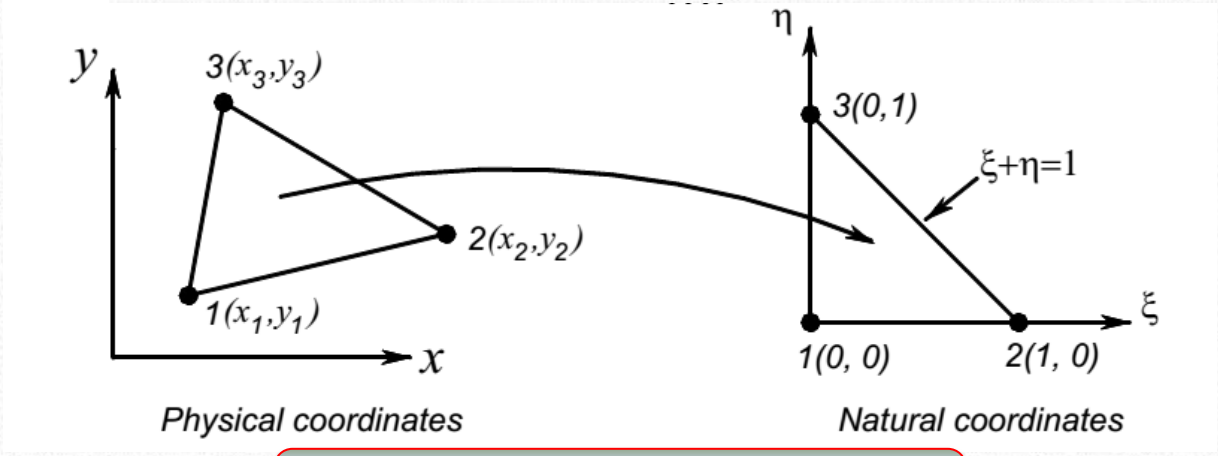
○ Biến dạng uốn $u = u_0 + z\beta_x \quad v = v_0 + z\beta_y$

$$\varepsilon_b = \begin{bmatrix} \dfrac{\partial u}{\partial x} \\ \dfrac{\partial v}{\partial y} \\ \dfrac{\partial u}{\partial y} + \dfrac{\partial v}{\partial x} \end{bmatrix} = \begin{bmatrix} \dfrac{\partial}{\partial x}(u_0 + z\beta_x) \\ \dfrac{\partial v}{\partial y}(v_0 + z\beta_y) \\ \dfrac{\partial}{\partial y}(u_0 + z\beta_x) + \dfrac{\partial}{\partial x}(v_0 + z\beta_y) \end{bmatrix} = z \begin{bmatrix} \dfrac{\partial \beta_x}{\partial x} \\ \dfrac{\partial \beta_y}{\partial y} \\ \dfrac{\partial \beta_x}{\partial y} + \dfrac{\partial \beta_y}{\partial x} \end{bmatrix} = z\kappa$$

$$\kappa = \begin{bmatrix} 0 & \dfrac{\partial}{\partial x} & 0 \\ 0 & 0 & \dfrac{\partial}{\partial y} \\ 0 & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial x} \end{bmatrix} \underbrace{\begin{bmatrix} w \\ \beta_x \\ \beta_y \end{bmatrix}}_{\mathbf{U}_\beta} = \begin{bmatrix} 0 & \dfrac{\partial}{\partial x} & 0 \\ 0 & 0 & \dfrac{\partial}{\partial y} \\ 0 & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial x} \end{bmatrix} \underbrace{\begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix}}_{\mathbf{N}_\beta} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta} = \underbrace{\begin{bmatrix} 0 & \dfrac{\partial N_i}{\partial x} & 0 \\ 0 & 0 & \dfrac{\partial N_i}{\partial y} \\ 0 & \dfrac{\partial N_i}{\partial y} & \dfrac{\partial N_i}{\partial y} \end{bmatrix}}_{\mathbf{B}^b} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta} = \mathbf{B}^b \mathbf{d}_\beta$$



➢ Thành lập công thức Bb



$$N_1 = 1 - \xi - \eta, \quad N_2 = \xi, \quad N_3 = \eta$$

$$\mathbf{B}^b = \begin{bmatrix} 0 & \frac{\partial N_i}{\partial x} & 0 \\ 0 & 0 & \frac{\partial N_i}{\partial y} \\ 0 & \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial y} \end{bmatrix} = \begin{bmatrix} 0 & \frac{\partial N_1}{\partial x} & 0 & 0 & \frac{\partial N_2}{\partial x} & 0 & 0 & \frac{\partial N_3}{\partial x} & 0 \\ 0 & 0 & \frac{\partial N_1}{\partial y} & 0 & 0 & \frac{\partial N_2}{\partial y} & 0 & 0 & \frac{\partial N_3}{\partial y} \\ 0 & \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial y} & 0 & \frac{\partial N_2}{\partial y} & \frac{\partial N_2}{\partial y} & 0 & \frac{\partial N_3}{\partial y} & \frac{\partial N_3}{\partial y} \end{bmatrix}$$

Nút 1 **Nút 2** **Nút 3**



Đổi biến trong tích phân kép và định thức Jacobian

Xét phép biến đổi $(x, y) \Rightarrow (\xi, \eta)$, với $x = x(\xi, \eta)$ và $y = y(\xi, \eta)$ là hàm có các đạo hàm riêng liên tục theo các biến ξ, η

Định thức cấp hai sâu đây hay còn gọi là định thức Jacobi cho phép biến đổi $(x,y) \Rightarrow (\xi, \eta)$ là

$$\mathbf{J} = \left[\frac{\partial(x,y)}{\partial(\xi,\eta)}\right] = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{bmatrix} \Rightarrow \mathbf{J}^{-1} = \left[\frac{\partial(\xi,\eta)}{\partial(x,y)}\right] = \begin{bmatrix} \frac{\partial \xi}{\partial x} & \frac{\partial \eta}{\partial x} \\ \frac{\partial \xi}{\partial y} & \frac{\partial \eta}{\partial y} \end{bmatrix}$$

$$\iint_{\Omega_{xy}} \mathrm{d}x\mathrm{d}y = \iint_{\Omega_{\xi\eta}} \det \mathbf{J}\, \mathrm{d}\eta\mathrm{d}\xi$$



➢ Thành lập công thức Bb

$$\frac{\partial N_i}{\partial x}=\frac{\partial N_i}{\partial \xi}\frac{\partial \xi}{\partial x}+\frac{\partial N_i}{\partial \eta}\frac{\partial \eta}{\partial x}=N_{i,\xi}\xi_{,x}+N_{i,\eta}\eta_{,x}$$

$$\frac{\partial N_i}{\partial y}=\frac{\partial N_i}{\partial \xi}\frac{\partial \xi}{\partial y}+\frac{\partial N_i}{\partial \eta}\frac{\partial \eta}{\partial y}=N_{i,\xi}\xi_{,y}+N_{i,\eta}\eta_{,y}$$

$$N_{1,\xi}=-1 \quad N_{2,\xi}=1 \quad N_{3,\xi}=0$$
$$N_{1,\eta}=-1 \quad N_{2,\eta}=0 \quad N_{3,\eta}=1$$

$$\left.\begin{aligned} x&=N_1x_1+N_2x_2+N_3x_3\\ y&=N_1y_1+N_2y_2+N_3y_3\end{aligned}\right\} \Rightarrow \begin{cases} x=(x_2-x_1)\xi+(x_3-x_1)\eta+x_1\\ y=(y_2-y_1)\xi+(y_3-y_1)\eta+y_1\end{cases}$$

$$\mathbf{J}=\left[\frac{\partial(x,y)}{\partial(\xi,\eta)}\right]=\begin{bmatrix} x_{,\xi} & x_{,\eta}\\ y_{,\xi} & y_{,\eta}\end{bmatrix} \Rightarrow \begin{bmatrix} \xi_{,x} & \xi_{,y}\\ \eta_{,x} & \eta_{,y}\end{bmatrix}=\mathbf{J}^{-1}$$

$$\Rightarrow \mathbf{K}_b^e=\int_{\Omega^e}\mathbf{B}_b^{\mathrm{T}}\mathbf{D}_b\mathbf{B}_b\,\mathrm{d}\Omega_e=\int_{\Omega^e}\mathbf{B}_b^{\mathrm{T}}\mathbf{D}_b\mathbf{B}_b\,\mathrm{d}x\mathrm{d}y=\int_0^1\int_0^{1-\xi}\mathbf{B}_b^{\mathrm{T}}\mathbf{D}_b\mathbf{B}_b\det\mathbf{J}\,\mathrm{d}\eta\mathrm{d}\xi$$



➢ Thành lập công thức Bb

Bảng 3.2: Các điểm tích phân Gauss và các trọng số tương ứng cho phần tử tam giác đẳng tham số trong miền 2D [14].

Số điểm Gauss $n_{\xi\eta}^{G}$	Tọa độ - ξ_j	Tọa độ - η_j	Trọng số w_j	Bậc chính xác
1 điểm	1/3	1/3	1/2	1
3 điểm	1/6 , 4/6 , 1/6	1/6 , 1/6 , 4/6	1/6 , 1/6 , 1/6	2
4 điểm	1/5,3/5,1/5,1/3	1/5,1/5,3/5,1/3	25/96,25/96,25/96,-9/32	3
6 điểm	0.0915762135098 0.8168475729805 0.0915762135098 0.4459484909160 0.1081030181681 0.4459484909160	0.0915762135098 0.0915762135098 0.8168475729805 0.4459484909160 0.4459484909160 0.1081030181681	0.05497587182766 0.05497587182766 0.05497587182766 0.11169079483901 0.11169079483901 0.11169079483901	4

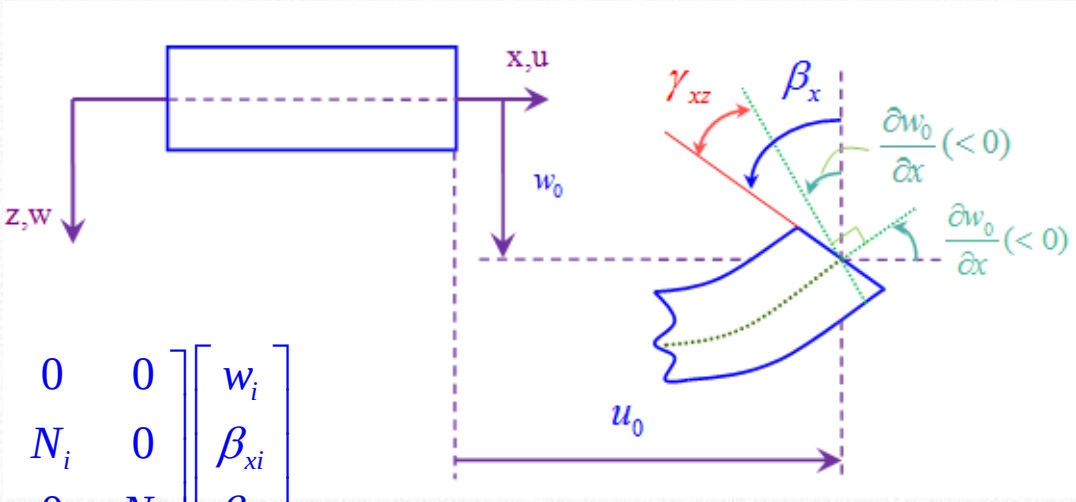


$$\mathbf{K}_b^e = \int_0^1 \left(\int_0^{1-\xi} \mathbf{B}_b^{\mathrm{T}} \mathbf{D}_b \mathbf{B}_b \det \mathbf{J} \, \mathrm{d}\eta \right) \mathrm{d}\xi = \sum_{i=1}^{n_{\xi\eta}^{G}} w_i \left[f(\xi_i, \eta_i) = \mathbf{B}_b^{\mathrm{T}} \mathbf{D}_b \mathbf{B}_b \det \mathbf{J}((\xi_i, \eta_i)) \right]$$

➢ Thành lập công thức Bs

○ Biến dạng cắt

$$\gamma = \begin{bmatrix} \gamma_{xz} \\ \gamma_{yz} \end{bmatrix} = \begin{bmatrix} w'_x + \beta_x \\ w'_y + \beta_y \end{bmatrix}$$

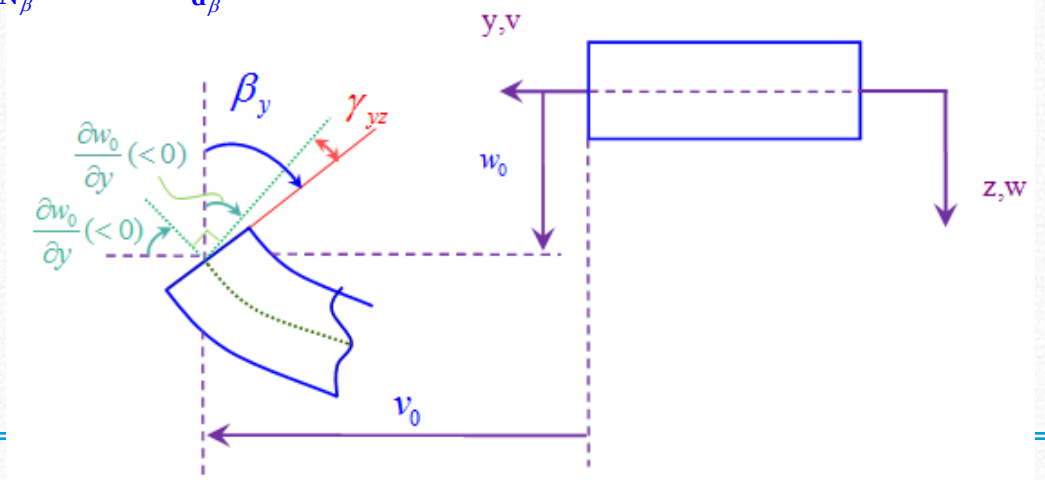


$$\gamma = \begin{bmatrix} \partial_{,x} & 1 & 0 \\ \partial_{,y} & 0 & 1 \end{bmatrix} \begin{bmatrix} w \\ \beta_x \\ \beta_y \end{bmatrix} = \begin{bmatrix} \partial_{,x} & 1 & 0 \\ \partial_{,y} & 0 & 1 \end{bmatrix} \underbrace{\begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix}}_{\mathbf{N}_\beta} \begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}$$

$\mathbf{U}_\beta$ $\qquad$ $\mathbf{d}_\beta$

$$= \begin{bmatrix} N_{i,x} & N_i & 0 \\ N_{i,y} & 0 & N_i \end{bmatrix} \begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix} = \mathbf{B}^s \mathbf{d}_\beta$$

$\mathbf{d}_\beta$





➢ Thành lập công thức Bs

○ Biến dạng cắt

$$\gamma = \begin{bmatrix} N_{i,x} & N_i & 0 \\ N_{i,y} & 0 & N_i \end{bmatrix} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta} = \underbrace{\begin{bmatrix} \overbrace{N_{1,x} \quad N_1 \quad 0}^{\text{Nút 1}} & \overbrace{N_{2,x} \quad N_2 \quad 0}^{\text{Nút 2}} & \overbrace{N_{3,x} \quad N_3 \quad 0}^{\text{Nút 3}} \\ N_{1,y} \quad 0 \quad N_1 & N_{2,y} \quad 0 \quad N_2 & N_{3,y} \quad 0 \quad N_3 \end{bmatrix}}_{\mathbf{B}^s} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta} = \mathbf{B}^s \mathbf{d}_\beta$$

$$\Rightarrow \quad \mathbf{K}_s^e = \int_{\Omega^e} \mathbf{B}_s^{\mathrm{T}} \mathbf{D}_s \mathbf{B}_s \, \mathrm{d}\Omega_e = \int_{\Omega^e} \mathbf{B}_b^{\mathrm{T}} \mathbf{D}_b \mathbf{B}_b \, \mathrm{d}x\mathrm{d}y = \int_0^1 \int_0^{1-\xi} \mathbf{B}_s^{\mathrm{T}} \mathbf{D}_s \mathbf{B}_s \det \mathbf{J} \, \mathrm{d}\eta \mathrm{d}\xi$$



$$\mathbf{K}_e = \mathbf{K}_b^e + \mathbf{K}_s^e$$



➢ Thành lập ma trận khối lượng

$$\mathbf{M}^e = \int_{\Omega_e} (\mathbf{N}^e)^T \mathbf{m} \mathbf{N}^e d\Omega$$

$$\mathbf{N}^e = \begin{bmatrix} N_1 & 0 & 0 & N_2 & 0 & 0 & N_3 & 0 & 0 \\ 0 & N_1 & 0 & 0 & N_2 & 0 & 0 & N_3 & 0 \\ 0 & 0 & N_1 & 0 & 0 & N_2 & 0 & 0 & N_3 \end{bmatrix}$$

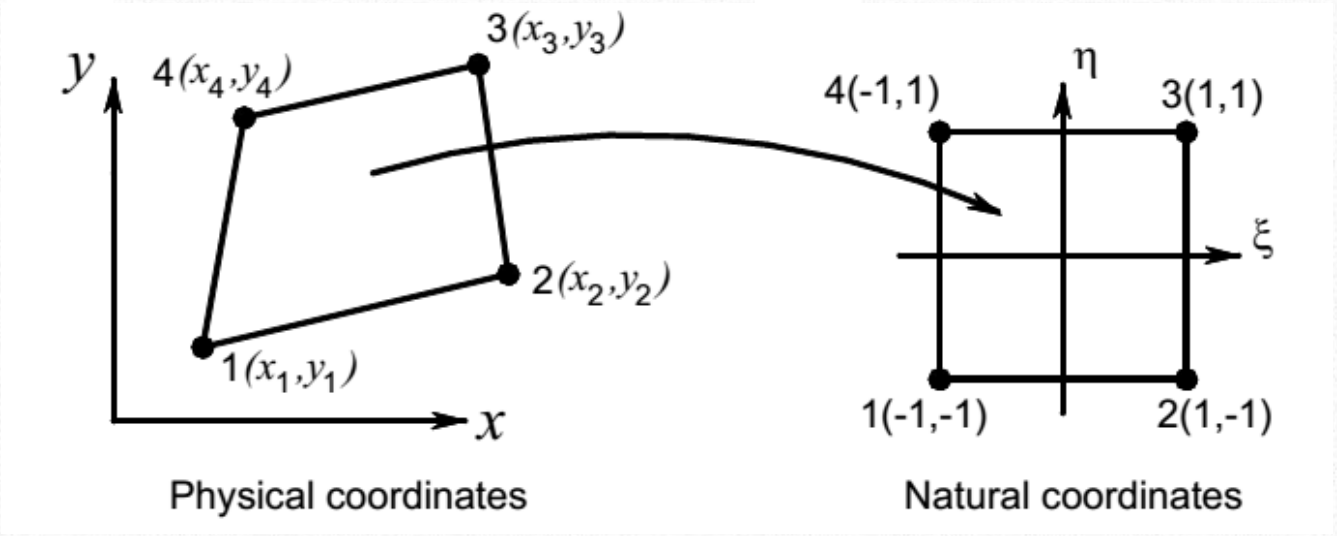
Nút 1 **Nút 2** **Nút 3**

$$\mathbf{m} = \rho \begin{bmatrix} t & 0 & 0 \\ 0 & \frac{t^3}{12} & 0 \\ 0 & 0 & \frac{t^3}{12} \end{bmatrix}$$

Để tính tích phân này ta sử dụng tích phân Gauss

FEM-Q4

➢ Thành lập công thức Bb



$$N_1 = \frac{1}{4}(1-\xi)(1-\eta);\quad N_2 = \frac{1}{4}(1+\xi)(1-\eta)$$

$$N_3 = \frac{1}{4}(1+\xi)(1+\eta);\quad N_4 = \frac{1}{4}(1-\xi)(1+\eta)$$

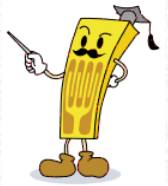
FEM-Q4



➢ Thành lập công thức Bb

$$\mathbf{B}^b = \begin{bmatrix} 0 & \frac{\partial N_i}{\partial x} & 0 \\ 0 & 0 & \frac{\partial N_i}{\partial y} \\ 0 & \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial y} \end{bmatrix} = \Bigg[\underbrace{\begin{bmatrix} 0 & \frac{\partial N_1}{\partial x} & 0 \\ 0 & 0 & \frac{\partial N_1}{\partial y} \\ 0 & \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial y} \end{bmatrix}}_{\text{Nút 1}} \underbrace{\begin{bmatrix} 0 & \frac{\partial N_2}{\partial x} & 0 \\ 0 & 0 & \frac{\partial N_2}{\partial y} \\ 0 & \frac{\partial N_2}{\partial y} & \frac{\partial N_2}{\partial y} \end{bmatrix}}_{\text{Nút 2}} \underbrace{\begin{bmatrix} 0 & \frac{\partial N_3}{\partial x} & 0 \\ 0 & 0 & \frac{\partial N_3}{\partial y} \\ 0 & \frac{\partial N_3}{\partial y} & \frac{\partial N_3}{\partial y} \end{bmatrix}}_{\text{Nút 3}} \underbrace{\begin{bmatrix} 0 & \frac{\partial N_4}{\partial x} & 0 \\ 0 & 0 & \frac{\partial N_4}{\partial y} \\ 0 & \frac{\partial N_4}{\partial y} & \frac{\partial N_4}{\partial y} \end{bmatrix}}_{\text{Nút 4}} \Bigg]$$

$$\Rightarrow \mathbf{K}_b^e = \int_{\Omega^e} \mathbf{B}_b^{\mathrm{T}} \mathbf{D}_b \mathbf{B}_b \, \mathrm{d}\Omega_e = \int_{\Omega^e} \mathbf{B}_b^{\mathrm{T}} \mathbf{D}_b \mathbf{B}_b \, \mathrm{d}x\mathrm{d}y = \int_{-1}^{1}\int_{-1}^{1} \mathbf{B}_b^{\mathrm{T}} \mathbf{D}_b \mathbf{B}_b \det \mathbf{J} \, \mathrm{d}\eta \mathrm{d}\xi$$





➢ Thành lập công thức Bs

○ Biến dạng cắt

$$\gamma = \begin{bmatrix} N_{i,x} & N_i & 0 \\ N_{i,y} & 0 & N_i \end{bmatrix} = \underbrace{\begin{bmatrix} N_{1,x} & N_1 & 0 & N_{2,x} & N_2 & 0 & N_{3,x} & N_3 & 0 & N_{4,x} & N_4 & 0 \\ N_{1,y} & 0 & N_1 & N_{2,y} & 0 & N_2 & N_{3,y} & 0 & N_3 & N_{4,y} & 0 & N_4 \end{bmatrix}}_{\mathbf{B}^s}$$

Nút 1 **Nút 2** **Nút 3** **Nút 4**

$$\mathbf{K}_s^e = \int_{\Omega^e} \mathbf{B}_s^{\mathrm{T}} \mathbf{D}_s \mathbf{B}_s \, \mathrm{d}\Omega_e = \int_{\Omega^e} \mathbf{B}_b^{\mathrm{T}} \mathbf{D}_b \mathbf{B}_b \, \mathrm{d}x\mathrm{d}y = \int_{-1}^{1}\int_{-1}^{1} \mathbf{B}_s^{\mathrm{T}} \mathbf{D}_s \mathbf{B}_s \det \mathbf{J} \, \mathrm{d}\eta \mathrm{d}\xi$$

$$\mathbf{K}_e = \mathbf{K}_b^e + \mathbf{K}_s^e$$



➢ Thành lập ma trận khối lượng

$$\mathbf{M}^e = \int_{\Omega_e} (\mathbf{N}^e)^T \mathbf{m} \mathbf{N}^e d\Omega$$

$$\mathbf{N}^e = \begin{bmatrix} N_1 & 0 & 0 & N_2 & 0 & 0 & N_3 & 0 & 0 & N_4 & 0 & 0 \\ 0 & N_1 & 0 & 0 & N_2 & 0 & 0 & N_3 & 0 & 0 & N_4 & 0 \\ 0 & 0 & N_1 & 0 & 0 & N_2 & 0 & 0 & N_3 & 0 & 0 & N_4 \end{bmatrix}$$

Nút 1 **Nút 2** **Nút 3** **Nút 4**

$$\mathbf{m} = \rho \begin{bmatrix} t & 0 & 0 \\ 0 & \frac{t^3}{12} & 0 \\ 0 & 0 & \frac{t^3}{12} \end{bmatrix}$$

Xử lý trong lập trình

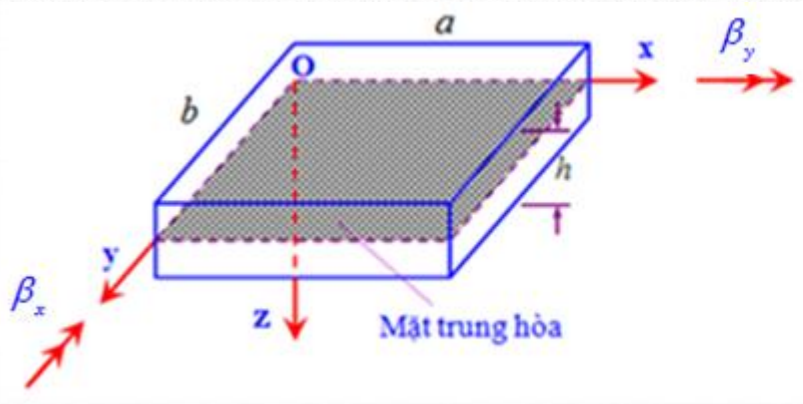


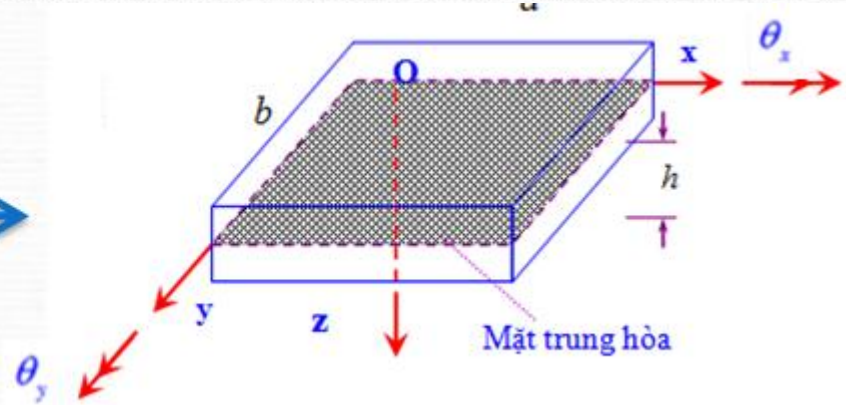
- Chuyển hệ trục tọa độ
  - Để thuận tiện trong các phép biến đổi, lập trình, ta đổi biến

$$\beta_y = \theta_x \quad ; \quad \beta_x = -\theta_y$$

$$\begin{aligned} u &= u_0 + z\beta_x \\ v &= v_0 + z\beta_y \\ w &= w_0 \end{aligned}$$

$$\begin{aligned} u &= u_0 - z\theta_y \\ v &= v_0 + z\theta_x \\ w &= w_0 \end{aligned}$$





Hệ tọa độ chính ***Hệ tọa độ chuyển đổi***

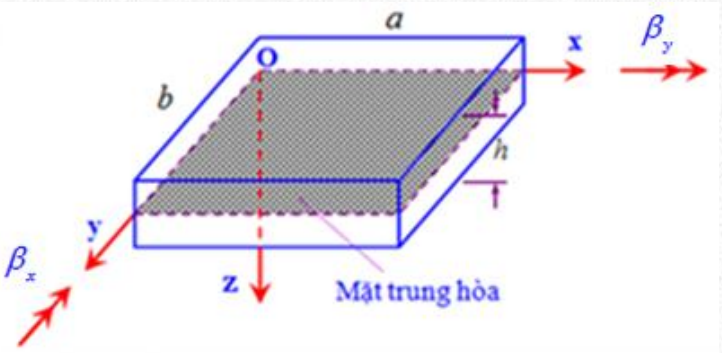
Xử lý trong lập trình

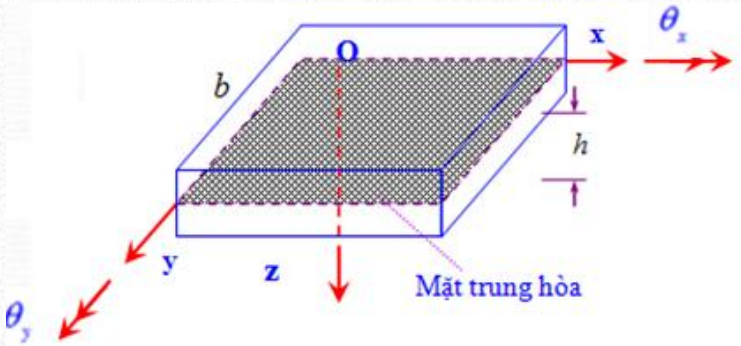


➢ Chuyển hệ trục tọa độ

○ Biến xấp xỉ theo β $\mathbf{U}_\beta = \mathbf{N}_\beta \mathbf{d}_\beta$

$$\mathbf{U}_\beta = \begin{bmatrix} w \\ \beta_x \\ \beta_y \end{bmatrix} = \begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix} \begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix} \begin{matrix} \Longrightarrow \text{Trục z} \\ \Longrightarrow \text{Trục y} \\ \Longrightarrow \text{Trục x} \end{matrix}$$





○ Biến xấp xỉ theo θ $\mathbf{U}_\theta = \mathbf{N}_\theta \mathbf{d}_\theta$

$$\mathbf{U}_\theta = \begin{bmatrix} w \\ \theta_x \\ \theta_y \end{bmatrix} = \begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix} \begin{bmatrix} w_i \\ \theta_{xi} \\ \theta_{yi} \end{bmatrix} = \begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix} \begin{bmatrix} w_i \\ \theta_{xi} \\ \theta_{yi} \end{bmatrix} \begin{matrix} \Longrightarrow \text{Trục z} \\ \Longrightarrow \text{Trục x} \\ \Longrightarrow \text{Trục y} \end{matrix}$$

Xử lý trong lập trình



➢ Chuyển hệ trục tọa độ

✓ Đổi biến $\beta_y = \theta_x \;\; ; \;\; \beta_x = -\theta_y$

$$\mathbf{U}_\beta = \begin{bmatrix} w \\ \beta_x \\ \beta_y \end{bmatrix} = \begin{bmatrix} w \\ -\theta_y \\ \theta_x \end{bmatrix} = \begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix} \begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix} \Leftrightarrow \mathbf{U}_\beta = \begin{bmatrix} w \\ \theta_y \\ \theta_x \end{bmatrix} = \begin{bmatrix} N_i & 0 & 0 \\ 0 & -N_i & 0 \\ 0 & 0 & N_i \end{bmatrix} \begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}$$

✓ Thay đổi vị trí của θ_x ; θ_y

$$\underbrace{\begin{bmatrix} w \\ \theta_x \\ \theta_y \end{bmatrix}}_{\mathbf{U}_\theta} = \begin{bmatrix} N_i & 0 & 0 \\ 0 & 0 & N_i \\ 0 & -N_i & 0 \end{bmatrix} \begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix} = \underbrace{\underbrace{\begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix}}_{\mathbf{N}_\beta} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}}_{\mathbf{T}}}_{\mathbf{N}_\theta} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta}$$

Xử lý trong lập trình



- Chuyển hệ trục tọa độ
  - ✓ Tất cả các công thức đều được thiết lập trên hệ tọa độ chính

$$\kappa_\beta = \begin{bmatrix} 0 & \frac{\partial}{\partial x} & 0 \\ 0 & 0 & \frac{\partial}{\partial y} \\ 0 & \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \underbrace{\begin{bmatrix} w \\ \beta_x \\ \beta_y \end{bmatrix}}_{\mathbf{U}_\beta} = \begin{bmatrix} 0 & \frac{\partial}{\partial x} & 0 \\ 0 & 0 & \frac{\partial}{\partial y} \\ 0 & \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \underbrace{\begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix}}_{\mathbf{N}_\beta} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta} = \begin{bmatrix} 0 & \frac{\partial N_i}{\partial x} & 0 \\ 0 & 0 & \frac{\partial N_i}{\partial y} \\ 0 & \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial x} \end{bmatrix} \begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix} = \mathbf{B}^b \mathbf{d}_\beta$$

✓ Để thuận tiện lập trình và xuất kết quả nên ta chuyển qua hệ tọa độ chuyển đổi

$$\kappa_\theta = \begin{bmatrix} 0 & \frac{\partial}{\partial x} & 0 \\ 0 & 0 & \frac{\partial}{\partial y} \\ 0 & \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \mathbf{U}_\theta = \begin{bmatrix} 0 & \frac{\partial}{\partial x} & 0 \\ 0 & 0 & \frac{\partial}{\partial y} \\ 0 & \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \underbrace{\begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix}}_{\mathbf{N}_\beta} \mathbf{T} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta} = \begin{bmatrix} 0 & \frac{\partial N_i}{\partial x} & 0 \\ 0 & 0 & \frac{\partial N_i}{\partial y} \\ 0 & \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial x} \end{bmatrix} \mathbf{T} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta} = \mathbf{B}^b \mathbf{T} \mathbf{d}_\beta$$

THANKS YOUR ATTENTION

